ART MEDIUM CLASSIFICATION

Image Processing Mini Project Report

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Abstract

This project investigates the automatic classification of artworks based on their artistic medium using image-processing, classical machine-learning, and deep-learning techniques.

A dataset containing 125 artwork images was manually collected and organized into five categories: Oil, Watercolour, Ink, Pastel, and Graphite/Pencil. Each category contains 25 images. The collected images were documented using metadata and validated before further processing.

The project was implemented in multiple stages. Initially, MATLAB was used for dataset validation and image preprocessing. The preprocessing pipeline standardized the images while preserving potentially useful visual information, including colour, texture, and strokes.

Classical feature extraction was subsequently performed using Harris Corner Detection and SIFT. The extracted features were evaluated using Support Vector Machine (SVM) and Cosine Similarity.

Deep-learning-based classification was also implemented using pretrained ResNet-18 and MViT V2-S architectures with modified classification layers for the five target classes.

Among the classical approaches, SIFT + SVM achieved the highest accuracy of 52.0%. Among the deep-learning models, ResNet-18 achieved the highest test accuracy of 52.63%, while MViT V2-S achieved 47.37%.

The experiments demonstrate the application of multiple image-analysis and classification techniques to art-medium recognition. They also highlight the challenges associated with limited dataset size, artistic variation, and source-related dataset bias.

1. Introduction

Artworks are created using different artistic mediums, including oil paints, watercolours, ink, pastels, and graphite.

Each medium can produce distinct visual characteristics. These characteristics may include:

- Colour distribution
- Texture
- Brush strokes
- Pencil marks
- Edge structures
- Shading patterns
- Intensity variations

Identifying the artistic medium from an artwork image can be challenging because different mediums may exhibit overlapping visual characteristics. Furthermore, artworks belonging to the same medium can vary significantly in style, subject, composition, and technique.

This project investigates the use of image processing and machine-learning techniques to automatically identify the artistic medium represented in an artwork image.

The work includes dataset preparation, classical feature extraction, machine-learning classification, deep-learning classification, and performance analysis.

2. Problem Statement

Given an image of an artwork, determine the most likely artistic medium used to create it.

The classification system considers five categories:

1. Oil
2. Watercolour
3. Ink
4. Pastel
5. Graphite/Pencil

The objective is to investigate whether visual information extracted from artwork images can be used to distinguish between these categories.

3. Objectives

The objectives of this project are:

1. To collect a real-world dataset of artwork images.
2. To organize the images into five artistic-medium categories.
3. To validate the collected dataset.
4. To preprocess images into a consistent representation.
5. To extract visual features using Harris Corner Detection and SIFT.
6. To evaluate the extracted features using SVM and Cosine Similarity.
7. To implement deep-learning-based classification using ResNet-18 and MViT V2-S.
8. To evaluate the classification performance.
9. To compare the experimental results and identify limitations.

4. Dataset Description

4.1 Dataset Collection

A dataset of 125 artwork images was manually collected from publicly available institutional art collections.

The sources include:

- The Metropolitan Museum of Art
- National Gallery of Art
- Art Institute of Chicago
- Cleveland Museum of Art

The images were manually organized into five categories.

4.2 Dataset Distribution

Category	**Number of Images**
Oil	25
Watercolour	25
Ink	25
Pastel	25
Graphite/Pencil	25
Total	**125**

The dataset contains an equal number of images in each category.

5. Dataset Validation

The raw dataset was validated using the MATLAB script check_dataset.m.

The script performs the following checks:

- Verifies the existence of all five category folders.
- Counts supported image files in each category.
- Checks whether images can be read successfully.
- Reports original image dimensions.
- Identifies missing, extra, or unreadable images.
- Displays an overall dataset summary.

Validation Result

Validation Item	Result
Expected images	125
Collected images	125
Images per category	25
Unreadable images	0
Dataset status	Valid

All 125 images were successfully validated before preprocessing.

6. Image Preprocessing

The collected artwork images originally had different dimensions and aspect ratios. Preprocessing was therefore performed to standardize their representation.

MATLAB was used to implement the preprocessing pipeline.

6.1 RGB Representation

Images were converted or maintained in a consistent three-channel RGB representation.

Colour information was preserved because it may be useful for distinguishing artistic mediums.

Grayscale conversion was avoided because it would remove colour information that could contribute to classification.

6.2 Pixel-Value Normalization

Images were converted to double precision using pixel values in the range [0,1] during processing.

Before saving, images were converted back to an 8-bit representation.

6.3 Resizing

The images were resized using bicubic interpolation while preserving their original aspect ratios.

This avoided unnecessary geometric distortion.

6.4 Center Padding

After resizing, the images were placed on a common 256 × 256 pixel canvas using center padding.

This ensured that every processed image had identical dimensions without arbitrarily cropping the original artwork.

6.5 Final Image Properties

Property	**Value**
Image dimensions	256 × 256 pixels
Colour representation	RGB
Normalization	[0,1] during processing
Interpolation	Bicubic
Aspect ratio	Preserved
Output format	PNG

6.6 Preprocessing Design Considerations

The preprocessing pipeline was designed to standardize the images while preserving visual characteristics relevant to art-medium recognition.

The following operations were intentionally avoided:

- Grayscale conversion
- Aggressive noise removal
- Histogram equalization

- Sharpening
- Strong contrast enhancement
- Arbitrary cropping

Such operations could modify colour distribution, texture, strokes, and shading patterns that may be useful for distinguishing artistic mediums.

The complete preprocessing procedure and visual examples are documented in Preprocessing_Documentation.pdf.

7. Classical Image Processing and Machine Learning

The preprocessed dataset was used for classical feature extraction and classification.

Two feature-extraction techniques were investigated:

1. Harris Corner Detection
2. SIFT (Scale-Invariant Feature Transform)

The extracted feature representations were evaluated using:

1. Support Vector Machine (SVM)
2. Cosine Similarity

The classical experiment used a training set of 100 images and a testing set of 25 images.

7.1 Classical Dataset Split

Dataset Split	**Number of Images**
Training	100
Testing	25
Total	**125**

The test set contained five images from each category.

8. Harris Corner Detection

Harris Corner Detection is a feature-detection technique that identifies points where image intensity changes significantly in multiple directions.

These points may correspond to:

- Corners
- Intersections
- Structural details
- Local patterns

For this project, Harris-based image features were extracted and converted into fixed-length feature vectors.

8.1 Harris Feature Representation

Each image was represented using 52 features.

The representation included:

- Four statistical features
- Twenty X-coordinate features
- Twenty Y-coordinate features
- Eight intensity histogram features

The resulting feature matrices were:

Dataset	Feature Shape
Training	100 × 52
Testing	25 × 52

These feature vectors were evaluated using SVM and Cosine Similarity.

9. SIFT Feature Extraction

SIFT stands for Scale-Invariant Feature Transform.

It detects distinctive local keypoints and generates descriptors representing the visual patterns surrounding those keypoints.

SIFT descriptors can capture local image characteristics such as:

- Edges
- Texture patterns
- Local intensity structures
- Stroke-related details

9.1 Bag of Visual Words

The number of SIFT keypoints can vary between images. Therefore, a fixed-length representation was required for classification.

A Bag of Visual Words (BoVW) approach was used.

The process involved:

1. Extracting SIFT descriptors from the training images.
2. Collecting the training descriptors.
3. Creating a visual vocabulary using clustering.
4. Representing each image using a histogram of visual-word occurrences.
5. Using the resulting fixed-length vectors for classification.

The training descriptors had the following shape:

(24916, 128)

The visual vocabulary contained 50 visual words.

Therefore, each image was represented using a 50-dimensional feature vector.

Dataset	Feature Shape
Training	100 × 50
Testing	25 × 50

The visual vocabulary was created using training-set descriptors.

10. Classical Classification Methods

10.1 Support Vector Machine

Support Vector Machine is a supervised machine-learning algorithm used for classification.

It identifies a decision boundary that separates different classes in the feature space.

For this project, SVM was applied to the feature representations obtained from:

- Harris Corner Detection
- SIFT

10.2 Cosine Similarity

Cosine Similarity measures the similarity between two feature vectors based on the cosine of the angle between them.

It is defined as:

$$CosineSimilarity(A,B) = (A \cdot B) / (\|A\| \|B\|)$$

For this experiment, cosine similarity was used to compare image feature representations with class-centroid representations and determine the most similar category.

11. Classical Machine Learning Results

The four combinations were evaluated using the same classical experiment's training and test partitions.

Feature Extraction	Classification Method	Accuracy
Harris Corner	SVM	24.0%
Harris Corner	Cosine Similarity	20.0%
SIFT	SVM	**52.0%**
SIFT	Cosine Similarity	48.0%

11.1 Best Classical Approach

The best-performing classical approach was:

SIFT + SVM – 52.0% accuracy.

This result indicates that SIFT-based feature representation was more effective than the Harris-based representation for this dataset.

SIFT + SVM also performed better than SIFT + Cosine Similarity.

11.2 Classification Report for SIFT + SVM

The classification report was:

Class	Precision	Recall	F1-Score	Support
Oil	0.75	0.60	0.67	5
Watercolour	0.50	0.40	0.44	5
Ink	0.38	0.60	0.46	5
Pastel	0.00	0.00	0.00	5
Graphite	0.62	1.00	0.77	5
Accuracy			**0.52**	**25**
Macro Average	0.45	0.52	0.47	25
Weighted Average	0.45	0.52	0.47	25

The model achieved its highest recall for Graphite, while Pastel was not correctly classified in this particular test set.

This demonstrates that overall accuracy alone does not fully describe classification performance across individual categories.

12. Deep Learning-Based Image Classification

Deep learning was implemented using PyTorch and Torchvision.

Two pretrained architectures were evaluated:

1. ResNet-18
2. MViT V2-S

The pretrained backbones were frozen, and the final classification layers were replaced with new layers corresponding to the five target categories.

12.1 Deep Learning Dataset Split

The dataset was divided into training, validation, and testing subsets.

Dataset Split	**Number of Images**
Training	87
Validation	19
Testing	19
Total	**125**

A stratified split was used to maintain class representation across the subsets.

Both models used the same dataset partitions and were evaluated on the same test images.

12.2 Image Input Preparation

The images were resized to 224 × 224 pixels for the deep-learning models.

ImageNet mean and standard deviation normalization were applied.

Training augmentation included:

- Random horizontal flipping
- Small random rotations

The validation and test transformations did not include random augmentation.

12.3 Transfer Learning

Pretrained model weights were used to initialize the architectures.

The pretrained backbone parameters were frozen, and only the replacement classification layer was trained.

This approach is known as transfer learning.

It allows the models to use previously learned visual representations while adapting the final classification layer to the five art-medium categories.

13. ResNet-18

ResNet-18 is a convolutional neural network architecture based on residual learning.

Residual connections allow information to pass through network layers more effectively.

For this experiment:

1. Pretrained ResNet-18 weights were loaded.
2. The original classification layer was replaced.
3. The replacement layer was configured for five output classes.
4. The pretrained backbone was frozen.
5. The new classification layer was trained using the artwork dataset.

The model produced class scores, which were converted into probabilities using Softmax during evaluation.

14. MViT V2-S

MViT V2-S is a multiscale vision transformer architecture.

It uses hierarchical visual representations and attention mechanisms to process image information at different scales.

For this experiment:

1. Pretrained MViT V2-S weights were loaded.
2. The final classification layer was replaced.
3. The replacement layer was configured for five output classes.
4. The pretrained backbone was frozen.
5. The new classification layer was trained using the artwork dataset.

A smaller batch size was used for MViT V2-S to accommodate its greater GPU memory requirements in the experimental setup.

The model was evaluated using the same test partition as ResNet-18.

15. Softmax Classification

The classification models produce raw output scores called logits.

Softmax converts these logits into probabilities across the five classes.

The Softmax function is:

$P(y_i) = e^{(z_i)} / \sum e^{(z_j)}$

where:

- (z_i) represents the score for a class.
- $(P(y_i))$ represents the predicted probability of that class.
- The sum of the probabilities across all classes is 1.

The class with the highest predicted probability is selected as the predicted art medium.

Cross-entropy loss was used during training with raw logits.

16. Deep Learning Results

The obtained test accuracies were:

Model	Test Accuracy
ResNet-18	52.63%
MViT V2-S	47.37%

16.1 ResNet-18

ResNet-18 achieved a test accuracy of 52.63%.

This corresponds to 10 correctly classified images out of the 19 test images.

16.2 MViT V2-S

MViT V2-S achieved a test accuracy of 47.37%.

This corresponds to 9 correctly classified images out of the 19 test images.

16.3 Result Interpretation

ResNet-18 performed slightly better than MViT V2-S in this experiment.

The difference corresponds to only one correctly classified test image.

The result may be influenced by:

- Limited dataset size
- Small number of training samples
- Model complexity
- Training configuration
- Batch-size differences
- Variations in artwork styles and visual characteristics

The result is specific to the current experiment and should not be interpreted as a general conclusion that ResNet-18 always outperforms MViT V2-S.

17. Overall Results and Comparison

The results from the classical and deep-learning experiments are summarized below.

Approach	Feature / Model	Method	Accuracy
Classical ML	Harris Corner	SVM	24.0%
Classical ML	Harris Corner	Cosine Similarity	20.0%
Classical ML	SIFT	SVM	52.0%
Classical ML	SIFT	Cosine Similarity	48.0%
Deep Learning	ResNet-18	Softmax Classification	52.63%
Deep Learning	MViT V2-S	Softmax Classification	47.37%

17.1 Overall Analysis

Among the classical approaches, SIFT + SVM achieved the highest accuracy of 52.0%.

Among the deep-learning models, ResNet-18 achieved the highest test accuracy of 52.63%.

These results are numerically close. However, the classical ML experiment used a 100/25 training/testing split, whereas the deep-learning experiment used an 87/19/19 training/validation/testing split.

Therefore, the two best accuracies are not a direct controlled comparison because they were obtained using different test sets.

The results demonstrate that classical feature-based approaches can remain competitive with deep-learning approaches on a small dataset, depending on the selected features and experimental configuration.

18. Limitations

18.1 Small Dataset Size

The dataset contains only 125 images across five categories.

Although this satisfies the mini-project dataset requirement, it is relatively small for learning robust distinctions between complex artistic-medium categories.

18.2 Artistic Variation

Artworks belonging to the same medium can differ significantly in:

- Subject
- Composition
- Artist
- Style
- Colour
- Historical period
- Technique

These variations can make classification challenging.

18.3 Similarity Between Categories

Different artistic mediums can exhibit overlapping visual characteristics.

For example, texture and shading patterns may appear similar across some categories.

18.4 Source-Related Dataset Bias

The images were collected from multiple institutional art collections. However, the distribution of sources is not uniform across all categories.

Consequently, source-related visual characteristics may introduce dataset bias.

A more source-diverse dataset would help investigate whether the models learn actual medium-related characteristics rather than source-specific patterns.

18.5 Small Test Set

The deep-learning test set contains only 19 images.

Therefore, one additional correct prediction changes the test accuracy by approximately 5.26 percentage points.

The classical test set contains 25 images, so one additional correct prediction changes its accuracy by 4 percentage points.

These small test sets limit the reliability of broad generalization claims.

19. Conclusion

This project implemented an end-to-end workflow for Art Medium Classification using image processing, classical machine learning, and deep learning.

The completed work includes:

- Manual dataset collection
- Dataset organization
- Metadata documentation
- Dataset validation
- Image preprocessing using MATLAB
- Harris Corner feature extraction
- SIFT feature extraction
- Bag of Visual Words representation
- SVM classification
- Cosine Similarity evaluation
- ResNet-18 classification
- MViT V2-S classification
- Softmax probability estimation
- Model evaluation and comparison

A dataset of 125 artwork images across five balanced categories was successfully collected and validated.

Among the classical approaches, SIFT + SVM achieved the highest accuracy of 52.0%.

Among the deep-learning models, ResNet-18 achieved the highest test accuracy of 52.63%, while MViT V2-S achieved 47.37%.

The results show that the effectiveness of a classification approach depends on the dataset, feature representation, architecture, and experimental configuration.

The limited dataset size and source distribution are important constraints when interpreting the results.

Overall, the project demonstrates the application of classical image-processing techniques and modern deep-learning architectures to the problem of artistic-medium classification.

20. Future Work

Possible future improvements include:

- Expanding the dataset with more artwork images.
- Increasing the number of images per category.
- Improving source diversity across all five categories.
- Including a wider variety of artists, styles, and compositions.
- Exploring additional classical feature-extraction techniques.
- Fine-tuning pretrained deep-learning backbones.
- Investigating other CNN and transformer architectures.
- Performing systematic hyperparameter tuning.
- Evaluating models on larger and more diverse test sets.
- Investigating class-specific performance and confusion patterns.

21. Tools and Technologies

Area	**Technologies**
Image preprocessing	MATLAB
Classical feature extraction	OpenCV
Classical classification	Scikit-learn

Numerical operations	NumPy
Data handling	Pandas
Deep learning	PyTorch
Model architectures	Torchvision
Execution environment	Google Colab
GPU acceleration	CUDA
Version control	Git and GitHub